MIL-E-1/1133C
22 October 1968
SUPERSEDING
MIL-E-1/1133B
28 November 1961

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 5AQP7

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Electrostatic deflection and focus, flat face

DIMENSIONS AND PIN CONNECTIONS:

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Rg	Zd	Ehk	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	Meg	Vdc	ft
Maximum:	6.9	0	400	1,650	4,400	1.5	1.0	+180	40,000
Minimum:	5.7	-200	---	---	---	---	---	-180	---
TEST CONDITIONS:	6.3	Adjust	---	Focus	3,500	---	---	---	---

GENERAL:

Qualification - Required (see note 1)

© denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 1							
5201	Voltage breakdown		0.65	II	---	---	---	---
5201	Voltage breakdown (electrostatic types)		0.65	II	---	---	---	---
5206	Gas "cross"	Light = 8 ftL (see notes 2 and 3)	0.65	II	---	---	---	---
5106	Screen and faceplate blemishes		0.65	II	---	---	---	---
5231	Spot position (electrostatic deflection)		0.65	II	---	---	16	mm
5251	Spot displacement (leakage)		0.65	II	Displ	---	10	mm
5241	Grid cutoff voltage		0.65	II	Eco	-47	-79	Vdc
5251	Grid No. 1 leakage		0.65	II	---	---	3	μAdc
5251	Anode No. 2 leakage		0.65	II	---	---	5	μAdc
---	Pattern distortion	See note 4	0.65	II	---	---	---	---
© 5223	Modulation	Light = 8 ftL (see note 2)	0.65	II	ΔEc	---	40	Vdc
	Quality conformance inspection, part 2							
1101	Secureness of base and base insert		6.5	S3	---	---	---	---
1301	Heater current		6.5	S3	If	540	660	mA
5201	Electrode currents (anode No. 1)	Ec1 = 0	6.5	S3	Ib1	-15	15	μAdc
5201	Electrode currents (anode No. 2)	Light = 8 ftL (see note 2)	6.5	S3	Ib2	---	350	μA
5201	Electrode currents (cathode)	Light = 8 ftL (see note 2)	6.5	S3	Ik	---	1,000	μAdc
5101	Base alinement (electrostatic types)	+1D2, pin No. 5	6.5	S3	---	---	---	---
5101	Neck and base alinement (electrostatic types)		6.5	S3	---	---	---	---
5101	Angle between traces		6.5	S3	---	89	91	Degrees
5216	Stray emission (conventional types)	Eb = 4,400 Vdc	6.5	S3	---	---	---	---
5221	Screen (P7 types)	See note 1	6.5	S3	---	---	---	---
5226	Line width A (electrostatic deflection)	Light = 8 ftL (see note 2)	6.5	S3	Width	---	0.75	mm
5226	Line width B (electrostatic deflection)	Light = 8 ftL (see note 2)	6.5	S3	Width	---	0.80	mm

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
5246	Focusing voltage at cutoff		6.5	S3	Eb1	---	420	Vdc
5246	Focusing voltage (zero bias)		6.5	S3	Eb1	0	---	Vdc
5248	Deflection factor (1D2)		6.5	S3	DF	57	69	Vdc/in.
5248	Deflection factor (3D4)		6.5	S3	DF	44	54	Vdc/in.
5251	Heater-cathode leakage		6.5	S3	---	---	10	μAdc
	Quality conformance inspection, part 3							
---	Life-test provisions	Group C; light = 8 ftL (see note 2); Eb2 = 4,400 Vdc; t = 500 hours (min)	---	---	---	---	---	---
---	Life-test end points	Line width A	---	---	Width	---	0.75	mm
		Line width B (see note 5)	---	---	Width	---	0.80	mm
		Modulation	---	---	ΔEc	---	40	Vdc
©	Periodic-check tests							
1141	Pressure		---	---	---	---	---	---
1002	Barometric pressure, reduced	130 mm Hg	---	---	---	---	---	---
5111	Vibration		---	---	Width	---	1.0	mm
1331	Direct-interelectrode capacitance	g1 to all	---	---	Cg1	---	6.9	pF
		k to all	---	---	Ck	---	5.0	pF
		D1 to D2	---	---	C1D2	---	4.5	pF
		D3 to D4	---	---	C3D4	---	1.6	pF
		D1 to all	---	---	CD1	---	9.3	pF
		D2 to all	---	---	CD2	---	9.3	pF
		D3 to all	---	---	CD3	---	6.3	pF
		D4 to all	---	---	CD4	---	6.3	pF
5101	Neck and bulb alinement (electrostatic types)		---	---	Dia	---	2.25	in.
5248	Deflection-factor uniformity		---	---	---	---	1	%
© 5216	Cathode illumination		---	---	---	---	---	---

NOTES:

1. For qualification tests the manufacturer shall submit a special sample having a split anode No. 2 to permit direct metering of screen current separate from anode No. 2 current. This sample shall be in addition to the regular qualification samples.

2. Measure in accordance with method 5221 using a Wratten No. 15 filter, or equivalent.

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NOTES: -Continued

3. This test to be performed at the conclusion of the holding period.

4. All portions of a raster pattern, adjusted so its widest points just touch the sides of a 3.672-inch square, will fall within the area bounded by the 3.672-inch square and an inscribed 3.528-inch square.

5. Measured at a distance from the center of the screen equal to one-third of the maximum bulb diameter.

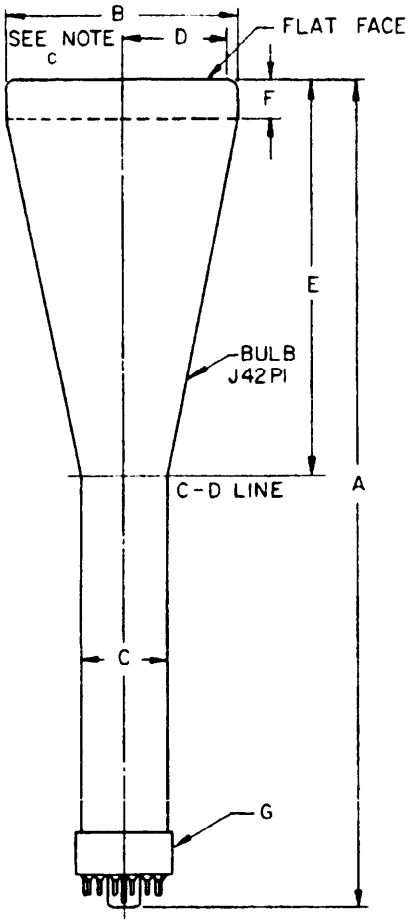
Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Air Force - 11
DSA - ES

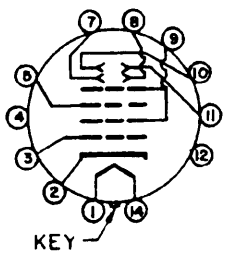
User activities:
Army - ME
Navy - AS, OS, CG
Air Force - 19

Preparing activity:
Navy - EC
(Project 5960-2225-5)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note a)	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	16.563 (420.70)	16.938 (430.23)
B	5.156 (130.96)	5.344 (135.74)
C	1.938 (49.23)	2.063 (52.40)
D	2.250 (57.15)	
Quality conformance inspection, part 3 (periodic check)		
G	Base: B12-37	
Reference dimensions (see note b)		
E	9.000 (228.60)	
F	.875 (22.23)	



BASING DIAGRAM

Pin Connections

Pin No.	Element
1	heater
2	cathode
3	grid No. 1
4	internal connection
5	focusing electrode
7	deflecting electrode D_3
8	deflecting electrode D_4
9	accelerator
10	deflecting electrode D_2
11	deflecting electrode D_1
12	none
14	heater

NOTES:
a. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
b. These dimensions are for information only and are not required for inspection purposes.
c. Minimum useful screen radius.

FIGURE 1. Outline drawing of electron tube type 5AQP7.

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